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Data Warehouses and Decision Support Systems (CO4031)

Assignment

"Applying Business Intelligence and Related Technologies to Hospital Performance Analytics"

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1 Introduction

1.1 Background and Motivation

In modern healthcare systems, hospitals generate massive volumes of data daily—from patient admissions and discharge records to treatment costs, departmental workloads, and clinical outcomes. However, much of this data remains underutilized due to fragmented storage systems, inconsistent formats, and the lack of analytical tools that can transform raw data into actionable insights. Hospital administrators often rely on manual reporting or basic spreadsheets, which limits their ability to identify trends, detect inefficiencies, and make timely decisions.

The emergence of Business Intelligence (BI) and Decision Support Systems (DSS), powered by Data Warehousing technologies, has fundamentally changed how organizations—including healthcare institutions—leverage their data assets. These technologies enable the integration, transformation, and visualization of complex datasets, providing decision-makers with an integrated and interactive view of organizational performance.

This project applies BI and related technologies to the domain of **Hospital Performance Analytics**, demonstrating how structured data pipelines and interactive dashboards can support evidence-based decision-making in healthcare management.

1.2 Key Concepts and Definitions

1.2.1 Business Intelligence (BI)

Business Intelligence (BI) refers to the technologies, practices, and strategies used to collect, integrate, analyze, and present business data. The primary goal of BI is to support better decision-making by transforming raw data into meaningful and useful information [4].

Key characteristics of BI include:

- **Data Integration:** Combining data from multiple sources into a unified view.
- **Descriptive and Diagnostic Analytics:** Answering questions such as "What happened?" and "Why did it happen?"
- **Interactive Reporting:** Dashboards, scorecards, and ad-hoc reports that allow users to explore data visually.
- **Historical and Real-time Analysis:** Supporting both retrospective trend analysis and up-to-date monitoring.

Example: A hospital uses a BI dashboard to monitor the Average Length of Stay (ALOS) across all departments. The dashboard shows that the Cardiology department has an ALOS of 7.2 days, significantly higher than the hospital average of 4.5 days. By drilling down, the hospital director discovers that a specific group of patients (age 65+) accounts for the majority of extended stays, prompting a review of post-operative care protocols.

1.2.2 Decision Support Systems (DSS)

A Decision Support System (DSS) is an interactive computer-based system designed to help decision-makers use data and models to solve semi-structured and unstructured problems [5]. Unlike transactional systems that focus on day-to-day operations, DSS focuses on providing analytical capabilities that support management-level decisions.

Key characteristics of DSS include:

- **Model-driven or Data-driven:** DSS can rely on mathematical/statistical models or on large data repositories (data warehouses).
- **Interactive and Flexible:** Users can query, filter, and manipulate data to explore different scenarios.
- **Support (not replace) Human Judgment:** DSS provides insights and alternatives, but the final decision rests with the human decision-maker.
- **Semi-structured Problem Solving:** Designed for problems where some aspects can be automated but human expertise is still required.

Example: A hospital manager uses a DSS to determine optimal staff allocation for the upcoming quarter. The system analyzes historical patient admission patterns (seasonal trends, peak hours) and suggests that the Emergency Department needs 20% more night-shift nurses during winter months due to increased respiratory cases.

1.2.3 Data Warehouse (DW)

A Data Warehouse is a centralized repository that stores integrated data from multiple sources, optimized for analytical querying and reporting rather than transaction processing [2]. It serves as the foundation for both BI and DSS by providing a consistent, clean, and historical data store.

Key characteristics of a Data Warehouse:

- **Subject-oriented:** Organized around key subjects (e.g., patients, treatments, departments) rather than applications.
- **Integrated:** Data from disparate sources is cleaned, transformed, and unified.
- **Time-variant:** Historical data is preserved to support trend analysis over time.
- **Non-volatile:** Once loaded, data is not modified—ensuring consistency for reporting.

1.2.4 Online Analytical Processing (OLAP)

OLAP is a category of software technology that enables analysts and managers to gain insight into data through fast, consistent, and interactive access to a wide variety of possible views of information [1]. OLAP supports operations such as:

- **Slice and Dice:** Viewing data from different perspectives (e.g., by department, by time period).
- **Drill-down / Roll-up:** Moving between levels of detail (e.g., from yearly totals to monthly breakdowns).
- **Pivoting:** Rotating the data axes to compare different dimensions.

1.3 Comparison: Business Intelligence vs. Decision Support Systems

Although BI and DSS are closely related and often overlap in practice, they differ in scope, focus, and application. Table 1 summarizes the key differences:

Table 1: Comparison between Business Intelligence and Decision Support Systems

Criteria	**Business Intelligence (BI)**	**Decision Support Systems (DSS)**
Primary Purpose	Monitor and report on organizational performance	Support specific decision-making processes
Focus	Descriptive analytics (What happened? Why?)	Prescriptive/predictive analytics (What should we do?)
Users	Broad range: executives, managers, analysts	Primarily managers and decision-makers
Data Approach	Data-driven (dashboards, reports, KPIs)	Model-driven or data-driven (what-if analysis, simulations)
Interactivity	Exploration through filters, drill-downs, slicers	Scenario analysis, parameter adjustment
Output	Reports, dashboards, alerts	Recommendations, alternative solutions
Scope	Enterprise-wide monitoring	Problem-specific decision support
Technology Base	DW, ETL, OLAP, Visualization tools	DW, Models, Optimization algorithms
Example Tools	Power BI, Tableau, Looker, QlikView	Custom analytical apps, Excel Solver

Relationship in this project: In this project, the system functions as both a BI platform and a data-driven DSS. The Power BI dashboards provide BI capabilities (monitoring KPIs, visualizing trends, comparing departments), while the analytical model and drill-down capabilities support DSS functions (identifying root causes, supporting resource allocation decisions). The Data Warehouse (Google BigQuery) serves as the shared foundation for both.

1.4 Related Technologies in This Project

- **ETL (Extract, Transform, Load):** The process of extracting data from source systems, transforming it into a consistent format, and loading it into the data warehouse. In this project, Python with Pandas is used for ETL.
- **Star Schema:** A dimensional modeling technique where a central fact table is surrounded by dimension tables. This design optimizes query performance for OLAP workloads.
- **Cloud Data Warehouse (Google BigQuery):** A serverless, highly scalable data ware-

house that supports SQL-based analytics on large datasets without infrastructure management.

- **Power BI:** A Microsoft BI tool that connects to data sources and provides interactive visualization, dashboard creation, and self-service analytics capabilities.

2 Business Problem

2.1 Problem Statement

Hospitals today face significant challenges in managing and analyzing their operational data effectively. Key issues include:

1. **Data Fragmentation:** Patient encounters, treatment costs, department workloads, and length-of-stay records are often stored in separate, disconnected systems (Electronic Health Records, billing systems). This fragmentation makes it difficult to obtain a holistic view of hospital performance.

2. **Lack of Analytical Capability:** Most hospital staff interact with transactional systems (OLTP) designed for day-to-day operations (admitting patients, billing). These systems are not designed for complex analytical queries such as "Which department has the highest cost-per-patient trend over the last 3 years?"

3. **Delayed Decision-Making:** Without integrated analytics, hospital managers must wait for periodic manual reports (often monthly or quarterly) to understand performance trends. By the time issues are identified, opportunities for timely intervention have passed.

4. **Inefficient Resource Allocation:** Without data-driven insights, decisions about budget allocation and departmental planning are often based on intuition rather than evidence, leading to suboptimal use of limited healthcare resources.

5. **Inability to Identify Root Causes:** When problems arise (e.g., increasing treatment costs, longer patient stays), managers lack the tools to quickly drill down into the data and identify contributing factors such as specific departments, patient demographics, or time periods.

2.2 Problem Context

Consider a mid-size hospital with 10+ departments, hundreds of daily patient encounters, and multiple attending physicians. The hospital director notices that overall treatment costs have increased by 15% year-over-year, but cannot determine:

- Which departments are driving the cost increase?
- Is the increase related to longer patient stays or higher per-day costs?
- Are specific patient demographics (age groups, genders) contributing disproportionately?
- How does the current year compare to previous years at the department level?

Without a BI/DSS system, answering these questions requires manual data extraction, spreadsheet manipulation, and significant time investment—resources that could be better spent on patient care.

2.3 Decision Makers and Stakeholders

The system is designed to serve multiple levels of hospital management, each with distinct analytical needs:

Table 2: Decision Makers and Their Analytical Needs

Stakeholder	**Role**	**Decision Needs**
Hospital Executives	Strategic planning, budget approval	Performance trends, YoY comparisons, cost efficiency, department benchmarking
Department Heads	Operational management of clinical departments	Department KPIs, workload distribution, staff performance, resource utilization
Administrative Staff	Day-to-day operations support	Patient record lookup, encounter details, scheduling support
QA Team	Monitoring care quality standards	Length of stay patterns, treatment cost outliers, encounter-level details

Primary Decision Makers: The Hospital Board of Directors and Department Heads are the primary target users. They need to:

- Monitor Key Performance Indicators (KPIs) in real-time or near-real-time.
- Compare performance across departments and time periods.
- Identify anomalies or concerning trends (e.g., rising costs, increasing stay durations).
- Make evidence-based decisions on resource allocation, staffing, and operational improvements.

2.4 Expected Goals

The project aims to achieve the following goals:

2.4.1 Primary Goals

1. **Build an integrated Data Warehouse:** Design and implement a star schema-based data warehouse on Google BigQuery that consolidates fragmented healthcare data into a single, query-optimized analytical store.
2. **Develop an interactive BI Dashboard:** Create a multi-page Power BI dashboard that provides hospital decision-makers with interactive visibility into KPIs including Total Encounters, Average Length of Stay, Total Treatment Cost, and Department Workload.
3. **Enable Multi-dimensional Analysis:** Support OLAP operations (drill-down, slice-and-dice, filtering) so that users can explore data across multiple dimensions: time, department, doctor, patient demographics.
4. **Support Root-Cause Investigation:** Provide analytical capabilities that allow managers to identify the underlying factors behind performance issues (e.g., which departments or patient groups drive cost increases).
5. **Demonstrate a Complete BI Pipeline:** Implement the full data pipeline from raw data extraction (ETL) through data warehousing to final visualization, demonstrating the practical application of BI/DSS concepts learned in the course.

2.4.2 Success Criteria

- The data warehouse successfully integrates data from multiple sources with consistent formatting.

- The dashboard loads and responds to user interactions within acceptable time frames.
- Decision-makers can answer key business questions (e.g., "Which department has the highest cost?") within 2–3 clicks.
- The system supports at least 4 levels of analytical depth: Overview → Department → Root-Cause → Detail Records.

3 Proposed solution

3.1 Solution Overview

To address the "Healthcare/Hospital Performance Analytics" business problem, our team proposes the development of a Business Intelligence (BI) system. This solution leverages modern data processing technologies to extract, transform, and model healthcare data from multiple sources. The primary objective is to provide hospital managers and decision-makers with an interactive analytical platform to monitor performance, analyze operational trends, and support data-driven decision-making.

Through this system, stakeholders can evaluate treatment costs, analyze patient length of stay, compare department workloads, and explore performance patterns across time and patient demographics.

3.2 System Architecture Diagram

The solution follows a modern data pipeline architecture, describing the flow of data from raw sources to analytical dashboards:

- **Data Sources:** The input data consists of two Kaggle-based healthcare CSV datasets. These datasets are transformed and standardized into a unified schema before being loaded into the data warehouse.
- **ETL Process (Python/Pandas):** Python scripts are used to clean the data (handling missing values and duplicates), standardize datetime formats, and generate derived attributes such as hospital stay duration.
- **Data Warehouse (Google BigQuery):** The processed data is stored in Google BigQuery as the central repository. A Star Schema is implemented to support efficient analytical queries.
- **BI & Analytics (Power BI):** Power BI connects to the final analytical view `vw_patient_encounter_analysis` in BigQuery to build dashboards and visualizations. This allows users to perform drill-down analysis and explore data interactively.

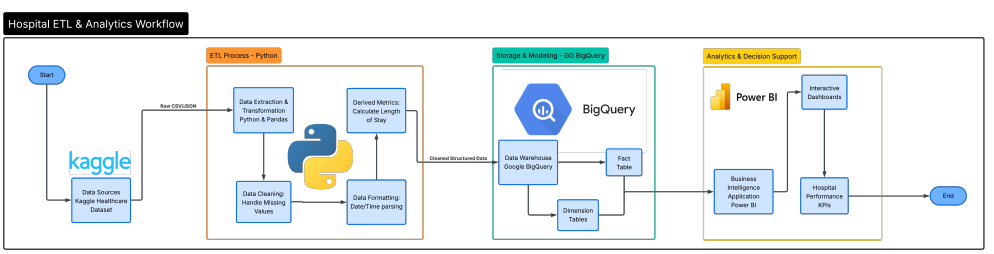


Figure 1: System Architecture Diagram

3.3 Data Modeling

The data warehouse is designed using a Star Schema following the Kimball dimensional modeling approach. This structure ensures efficient querying and intuitive data exploration by separating facts and dimensions.

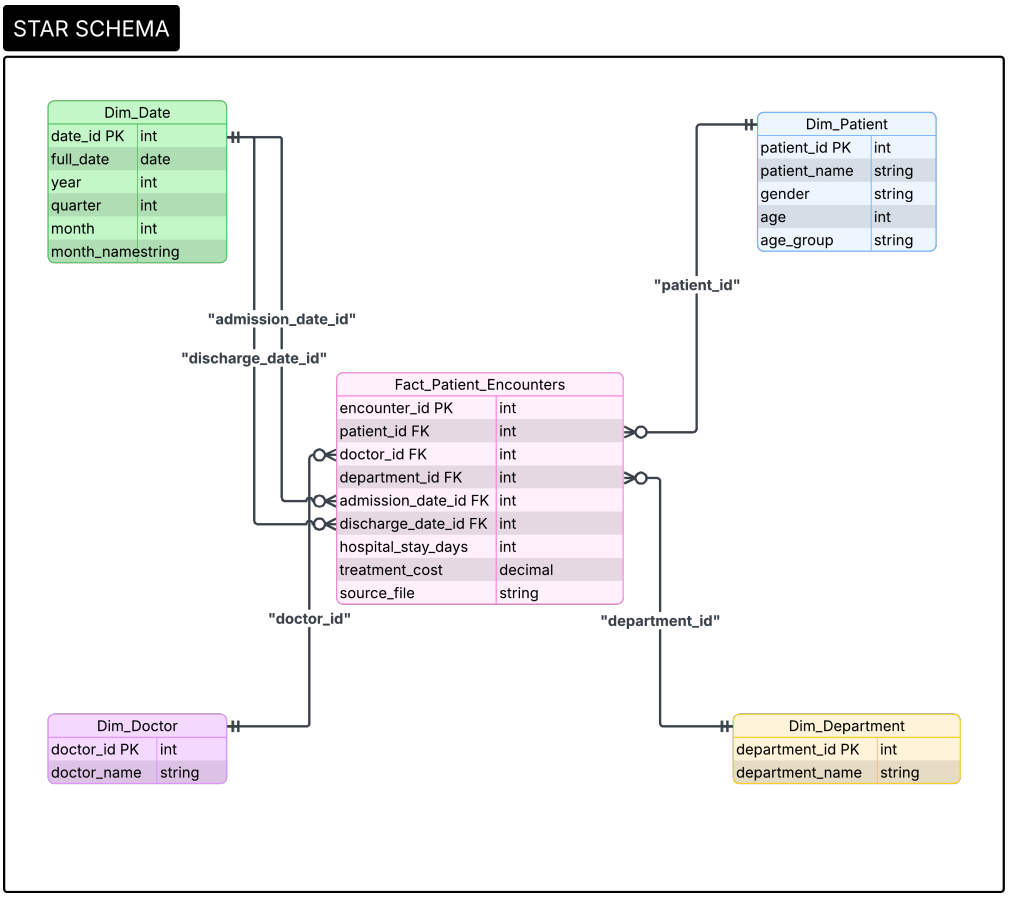


Figure 2: Data Modeling Star Schema

1. Fact Table: Fact_Patient_Encounters

This table records each patient encounter at the grain of one row per admission-discharge event.

- Grain: One row per patient encounter
- Primary Key:
  - Encounter_ID
- Foreign Keys:
  - Patient_ID, Doctor_ID, Department_ID
  - Admission_Date_ID, Discharge_Date_ID
- Measures:
  - hospital_stay_days
  - treatment_cost
  - source_file

2. Dimension Tables

- **Dim_Patient:**
  - patient_id, patient_name, gender, age, age_group
- **Dim_Doctor:**
  - doctor_id, doctor_name
- **Dim_Department:**
  - department_id, department_name
- **Dim_Date:**
  - date_id, full_date, year, quarter, month, month_name

(Note: Dim_Date is used as a role-playing dimension for both admission and discharge dates.)

3.4 Business Intelligence and Related Technologies

Power BI is used to build interactive dashboards that support the following analytical capabilities:

- **KPI Monitoring:** Track key indicators such as total encounters, average length of stay, and treatment cost.
- **Department Comparison:** Compare workload and performance across hospital departments.
- **Root-Cause Exploration:** Identify performance patterns by filtering data based on dimensions such as age group or department.
- **Drill-down Analysis:** Allow users to navigate from aggregated metrics to detailed encounter-level records.

3.5 Dimensional Bus Matrix

The Dimensional Bus Matrix defines the relationship between the primary business process and shared dimensions.

Business Process	Patient Encounters
Dim_Date	X
Dim_Patient	X
Dim_Doctor	X
Dim_Department	X

Table 3: Dimensional Bus Matrix

3.6 Analytical Model

The analytical model is implemented in Power BI based on the Kimball dimensional modeling approach. The model follows a star schema design, where the central fact table is connected to multiple dimension tables through well-defined relationships.

3.6.1 Multi-Dimensional Relationships

Role-Playing Dimension: The Dim_Date table is reused for both Admission Date and Discharge Date. The Admission Date is set as the active relationship to support primary time-

based analysis.

Star Schema Principles: Relationships between fact and dimension tables are configured as single-directional to ensure efficient query performance and avoid ambiguity in filtering.

3.6.2 Analytical Hierarchies (Drill-down Paths)

- Temporal Hierarchy: Year → Quarter → Month → Date
- Organizational Hierarchy: Department → Doctor
- Patient Demographic Hierarchy: Age_Group → Gender

3.6.3 Essential Calculated Measures

The following key performance indicators are implemented in the analytical model:

Total Encounters:

$$\text{Total Encounters} = \text{COUNT}(Encounter_ID)$$

Patient Count:

$$\text{Patient Count} = \text{DISTINCTCOUNT}(patient_id)$$

Average Length of Stay (ALOS):

$$\text{ALOS} = \frac{\sum \text{hospital_stay_days}}{\text{Total Encounters}}$$

Total Treatment Cost:

$$\text{Total Treatment Cost} = \sum \text{treatment_cost}$$

Average Treatment Cost:

$$\text{Average Treatment Cost} = \frac{\sum \text{treatment_cost}}{\text{Total Encounters}}$$

Department Workload:

$$\text{Department Workload} = \text{COUNT}(Encounter_ID \text{ by Department})$$

4 Extract - Transform - Load (ETL)

4.1 Objective

The primary objective of this process is to extract and transform clinical data into a standardized reporting format. The final dataset includes comprehensive Patient Demographics (Name, Gender, and Age) and Provider Details (Doctor Name and Department). Additionally, the report captures critical administrative and financial metrics, specifically the Date of Admission, Discharge Date, Total Hospital Stay (Days), and the final Treatment Cost. The following columns are included in the formatted dataset:

1. Patient Name
2. Age
3. Gender
4. Doctor Name
5. Department
6. Date of Admission
7. Discharge Date
8. Hospital Stay (Days)
9. Treatment Cost

4.2 Data Source & Information

4.2.1 **Healthcare Dataset** by Prasad Patil

- **URL:** https://www.kaggle.com/datasets/prasad22/healthcare-dataset
- **Shape:** 55500 rows x 15 columns
- **Dataset Information:**
    1. **Name**: Patient's name.
    2. **Age**: Patient's age at admission (in years).
    3. **Gender**: Male or Female.

4. **Blood Type**: Patient's blood group (e.g., A+, O-).
5. **Medical Condition**: Primary diagnosis (e.g., Diabetes, Hypertension).
6. **Date of Admission**: Date the patient was admitted.
7. **Doctor**: Attending physician.
8. **Hospital**: Admitting healthcare facility.
9. **Insurance Provider**: Patient's insurance company (e.g., Aetna, Medicare).
10. **Billing Amount**: Total cost billed for services.
11. **Room Number**: Assigned hospital room.
12. **Admission Type**: Emergency, Elective, or Urgent.
13. **Discharge Date**: Date the patient was discharged.
14. **Medication**: Prescribed medication (e.g., Aspirin, Lipitor).
15. **Test Results**: Outcome of medical tests (Normal, Abnormal, or Inconclusive).

4.2.2 Hospital Patient Treatment Dataset by Slidescope

- **URL:** https://www.kaggle.com/datasets/slidescope/hospital-patient-treatment-dataset
- **Shape:** 200 rows x 9 columns
- **Dataset Information:**

1. **Patient ID**: Unique identifier for the patient.
2. **Department**: Hospital department providing care.
3. **Treatment Type**: Specific medical treatment administered.
4. **Doctor Name**: Attending physician.
5. **Gender**: Patient's gender.
6. **Age**: Patient's age (in years).
7. **Treatment Cost**: Total billed cost of the treatment.
8. **Hospital Stay (Days)**: Number of days the patient remained admitted.
9. **Recovery Score**: Metric indicating the patient's recovery progress.

4.3 Extract - Data Extraction

The extraction phase focuses on retrieving raw data from specific Kaggle sources and loading it into a staging environment for processing. This step ensures that data integrity is maintained before any transformations occur. The datasets can be directly downloaded from the Kaggle website and loaded into a working Python environment using the `pandas` library.

4.4 Transform

4.4.1 Healthcare Dataset by Prasad Patil

- **Data Cleaning:**
  - **Name:** In the original dataset, the names were inconsistently capitalized. Therefore, we need to convert them to title case, capitalizing only the first letter of each word and lowercasing the rest (e.g., Bobby JacksOn → Bobby Jackson).
  - **Doctor:** In the original dataset, doctor names are displayed with their full first and last names. For the transformation, the first name is abbreviated to its initial letter followed by a period, while the last name remains fully spelled out (e.g., Matthew Smith → M. Smith).
  - **Billing Amount:** In the original dataset, the cost values had multiple decimal places. During the transformation, we rounded them to two decimal places (e.g., 18856.281305978155 → 18856.28).
- **Data Processing:**
  - **Hospital Stay (Days):** This column did not originally exist in the dataset. However, because the `Date of Admission` and `Discharge Date` were provided, we calculated the duration using Python's `datetime` library (e.g., DoA: 2024-01-31, DD: 2024-02-02 → HS: 2).
  - **Department:** Originally, this column did not exist. However, since each doctor belongs to a specific department, we can easily deduce which department the patient visited.
  - **Drop Columns:** Unnecessary columns were removed to keep the data clean and readable. The dropped columns include: Blood Type, Medical Condition, Hospital, Insurance Provider, Room Number, Admission Type, Medication, and Test Results.

  - **Rename Columns:** To maintain alignment with other datasets, the following columns have been renamed: `Name` to `Patient Name`, `Doctor` to `Doctor Name`, and `Billing Amount` to `Treatment Cost`.

- **Save as New File:** The newly processing dataframe is saved as `raw_healthcare_dataset.csv` to distinguish itself from the original dataset.

4.4.2 Hospital Patient Treatment Dataset by Slidescope

- **Data Cleaning:**
  - **Doctor:** The original dataset provided a shortened first name and a full last name with a "Dr." prefix. Therefore, during the transformation process, the name was formatted to its initial letter followed by a period, and the prefix was removed, while the last name remained fully spelled out (e.g., Dr. L. Khan → L. Khan).
- **Data Processing:**
  - **Patient Name:** With the `Patient ID`, names can be inferred using the existing database that stores patient information. However, because this is a public dataset and given the nature of the assignment, we have assumed access to the database and assigned names randomly instead.
  - **Date of Admission & Discharge Date:** This information is not mentioned; therefore, they are added as an empty string.
  - **Drop Columns:** Unnecessary columns were removed to keep the data clean and readable. The dropped columns include: Patient ID, Treatment Type, and Recovery Score.
- **Save as New File:** The newly processing dataframe is saved as `raw_treatment_dataset.csv` to distinguish itself from the original dataset.

4.5 Load

After the transformation process, the two formatted CSV files were loaded into Google BigQuery for data warehousing and analytical processing. Since both datasets were transformed into the same structure, they were first uploaded into BigQuery as two separate raw tables:

- `raw_healthcare_dataset`

- raw_treatment_dataset

The two raw tables were then combined using a UNION ALL operation to create a unified cleaned table named clean_patient_encounters. This cleaned table serves as the standardized encounter-level dataset for further modeling and analysis.

The load process follows this pipeline:

Transformed CSV files → BigQuery raw tables → clean_patient_encounters → fact and dimension tables → vw_patient_encounter_analysis → Power BI dashboard

From the cleaned table, the data was organized into a star schema structure. The central fact table, fact_patient_encounters, stores encounter-level measures such as hospital stay days and treatment cost. Several dimension tables were also created, including dim_patient, dim_doctor, dim_department, and dim_date.

Finally, a BigQuery analytical view named vw_patient_encounter_analysis was created by joining the fact table with the related dimension tables. This view acts as the final data source for the Power BI dashboard. Instead of connecting Power BI directly to raw tables, the dashboard connects to this cleaned and structured view to ensure consistency, easier visualization, and better analytical performance.

5 Application Development

5.1 Overview of the Developed Application

The application developed in this project is an interactive Power BI dashboard for hospital performance analytics. The dashboard is designed to help hospital decision makers monitor hospital workload, treatment cost, length-of-stay performance, and department-level patterns based on the available healthcare dataset.

Instead of using raw CSV files directly, Power BI connects to the final BigQuery analytical view, `vw_patient_encounter_analysis`. This view provides the cleaned and structured data needed for visualization, including patient information, doctor information, department information, date attributes, hospital stay duration, and treatment cost. The detailed ETL process and data warehouse structure are explained in the ETL and data modeling sections, while this section focuses on the developed dashboard application and its analytical functions.

The dashboard includes four main pages:

- Hospital Overview
- Department Performance
- Root-Cause Analysis
- Drill-down Records

Together, these pages support a decision-making flow from high-level monitoring to department comparison, pattern exploration, and detailed record inspection.

5.2 Main KPIs Used in the Dashboard

The dashboard uses KPIs that are directly supported by the available dataset. These KPIs focus on hospital workload, treatment cost, and length-of-stay performance.

The main KPIs include: Total Encounters, Patient Count, Average Length of Stay, Total Treatment Cost, Average Treatment Cost, Department Workload

These KPIs were selected because they can be calculated from the available fields, including patient records, department, admission date, discharge date, hospital stay duration, and treatment cost.

Metrics such as Readmission Rate, Bed Occupancy Rate, and Patient Satisfaction Score are not included in the implemented dashboard because the current dataset does not contain the required fields to calculate them reliably.

5.3 Dashboard Page 1: Hospital Overview

The Hospital Overview page provides a high-level summary of hospital performance. It is mainly designed for hospital executives and managers who need a quick view of overall hospital activity.

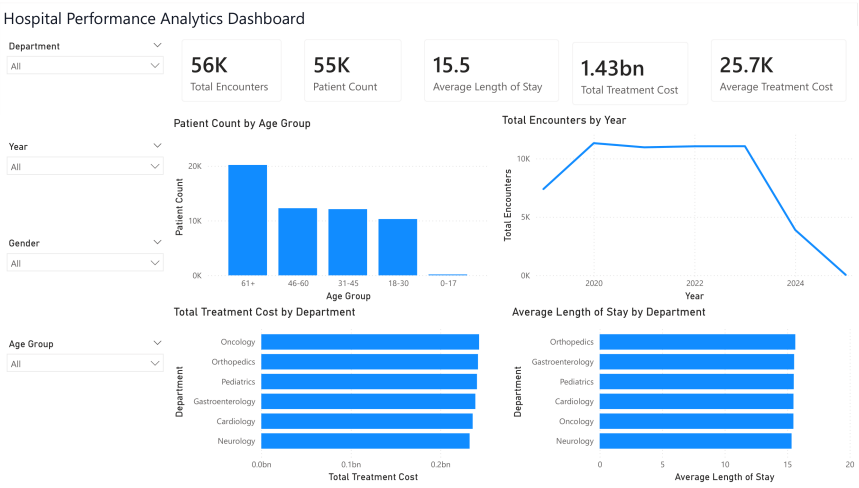


Figure 3: Hospital Overview page

This page includes five KPI cards:

- Total Encounters
- Patient Count
- Average Length of Stay
- Total Treatment Cost
- Average Treatment Cost

In addition, the page contains visualizations such as:

- Patient Count by Age Group

- Total Encounters by Year
- Total Treatment Cost by Department
- Average Length of Stay by Department

These visuals allow users to quickly identify overall workload trends, cost distribution across departments, and length-of-stay patterns. The overview page acts as the starting point of the dashboard because it summarizes the most important performance indicators in one place.

5.4 Dashboard Page 2: Department Performance

The Department Performance page focuses on comparing hospital departments. This page is useful for department heads and hospital managers who need to evaluate department-level workload, cost, and stay duration.

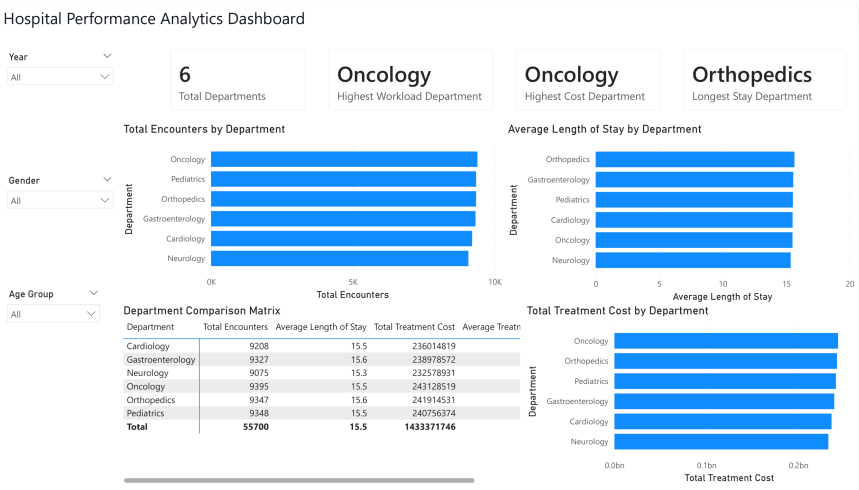


Figure 4: Department Performance page

The page includes department-level summary cards such as:

- Total Departments
- Highest Workload Department
- Highest Cost Department
- Longest Stay Department

It also includes visualizations for:

- Total Encounters by Department
- Average Length of Stay by Department
- Total Treatment Cost by Department
- Department Comparison Matrix

The Department Comparison Matrix allows users to compare departments across multiple indicators, including total encounters, average length of stay, total treatment cost, and average treatment cost. This page helps decision makers identify which departments may require closer attention due to higher workload, higher cost, or longer patient stays.

5.5 Dashboard Page 3: Root-Cause Analysis

The Root-Cause Analysis page is designed to help users explore possible patterns behind workload and treatment cost differences. Since the current dataset does not support readmission or bed occupancy analysis, this page focuses on dimensions that are available and reliable, such as age group, gender, department, and time period.

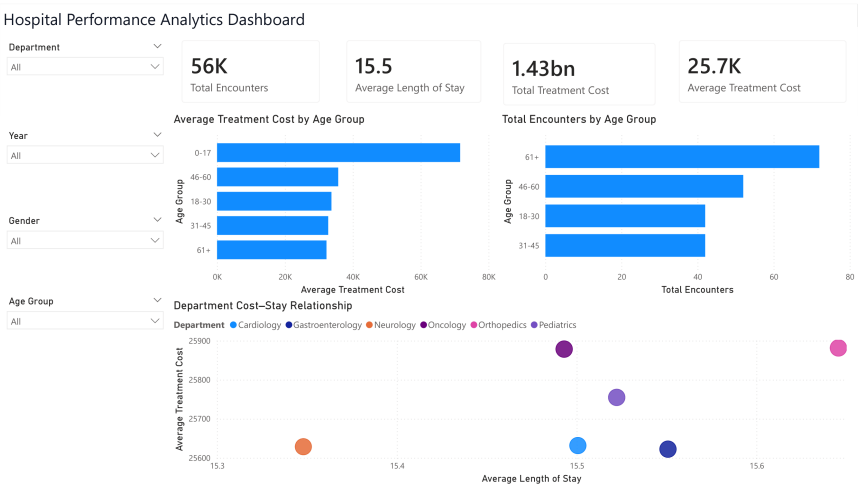


Figure 5: Root-Cause Analysis page

This page includes visualizations such as:

- Average Treatment Cost by Age Group

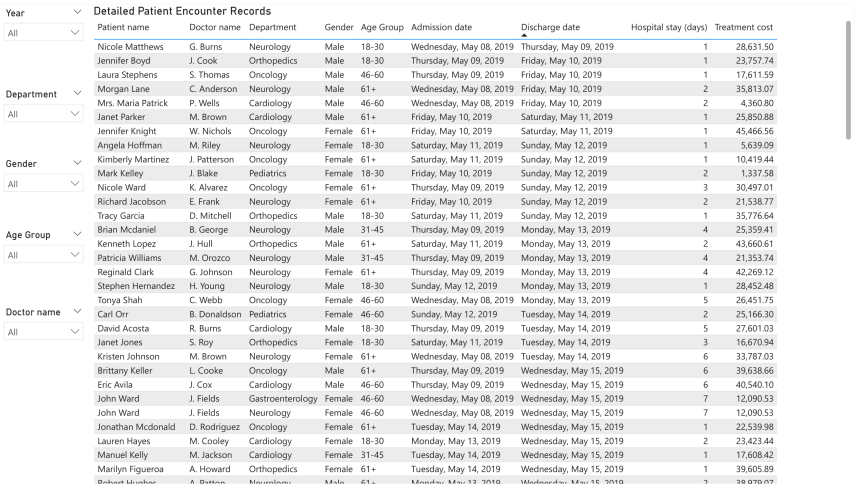
- Total Encounters by Age Group
- Department Cost–Stay Relationship

The purpose of this page is to help users identify whether certain patient groups or departments are associated with higher treatment costs or greater workload. For example, managers can observe which age groups contribute more encounters or which departments show higher average treatment cost in relation to length of stay.

This page supports root-cause exploration at an aggregated level before users move to detailed patient encounter records.

5.6 Dashboard Page 4: Drill-down Records

The Drill-down Records page provides detailed patient encounter information. This page is designed for users who need to inspect individual records after identifying a potential pattern or issue from the previous pages.



Patient name	Doctor name	Department	Gender	Age Group	Admission date	Discharge date	Hospital stay (days)	Treatment cost
Nicole Matthews	G. Burns	Neurology	Male	18-30	Wednesday, May 08, 2019	Thursday, May 09, 2019	1	28,631.50
Jennifer Boyd	J. Cook	Orthopedics	Male	18-30	Thursday, May 09, 2019	Friday, May 10, 2019	1	23,757.74
Laura Stephens	S. Thomas	Oncology	Male	46-60	Thursday, May 09, 2019	Friday, May 10, 2019	1	17,611.59
Morgan Lane	C. Anderson	Neurology	Male	61+	Wednesday, May 08, 2019	Friday, May 10, 2019	2	35,813.07
Mrs. Maria Patrick	P. Wells	Cardiology	Male	46-60	Wednesday, May 08, 2019	Friday, May 10, 2019	2	4,360.80
Janet Parker	M. Brown	Cardiology	Male	61+	Friday, May 10, 2019	Saturday, May 11, 2019	1	25,850.88
Jennifer Knight	W. Nichols	Oncology	Female	61+	Friday, May 10, 2019	Saturday, May 11, 2019	1	45,466.56
Angela Hoffman	M. Riley	Neurology	Female	18-30	Saturday, May 11, 2019	Sunday, May 12, 2019	1	5,639.09
Kimberly Martinez	J. Patterson	Oncology	Female	61+	Saturday, May 11, 2019	Sunday, May 12, 2019	1	10,419.44
Mark Kelley	J. Blake	Pediatrics	Female	18-30	Friday, May 10, 2019	Sunday, May 12, 2019	2	1,337.58
Nicole Ward	K. Alvarez	Oncology	Female	61+	Thursday, May 09, 2019	Sunday, May 12, 2019	3	30,497.01
Richard Jacobson	E. Frank	Neurology	Female	61+	Friday, May 10, 2019	Sunday, May 12, 2019	2	21,538.77
Tracy Garcia	D. Mitchell	Orthopedics	Male	18-30	Saturday, May 11, 2019	Sunday, May 12, 2019	1	35,776.64
Brian Mcdaniel	B. George	Neurology	Male	31-45	Thursday, May 09, 2019	Monday, May 13, 2019	4	25,359.41
Kenneth Lopez	J. Hull	Orthopedics	Male	61+	Saturday, May 11, 2019	Monday, May 13, 2019	2	43,660.61
Patricia Williams	M. Orozco	Neurology	Male	31-45	Thursday, May 09, 2019	Monday, May 13, 2019	4	21,353.74
Reginald Clark	G. Johnson	Neurology	Female	61+	Thursday, May 09, 2019	Monday, May 13, 2019	4	42,269.12
Stephen Hernandez	H. Young	Neurology	Male	18-30	Sunday, May 12, 2019	Monday, May 13, 2019	1	28,452.48
Tonya Shah	C. Webb	Oncology	Female	46-60	Wednesday, May 08, 2019	Monday, May 13, 2019	5	26,451.75
Carl Orr	B. Donaldson	Pediatrics	Female	46-60	Sunday, May 12, 2019	Tuesday, May 14, 2019	2	25,166.30
David Acosta	R. Burns	Cardiology	Male	18-30	Thursday, May 09, 2019	Tuesday, May 14, 2019	5	27,601.03
Janet Jones	S. Roy	Orthopedics	Female	18-30	Saturday, May 11, 2019	Tuesday, May 14, 2019	3	16,670.94
Kristen Johnson	M. Brown	Neurology	Female	61+	Wednesday, May 08, 2019	Tuesday, May 14, 2019	6	33,787.03
Brittany Keller	L. Cooke	Oncology	Male	61+	Thursday, May 09, 2019	Wednesday, May 15, 2019	6	39,638.66
Eric Avila	J. Cox	Cardiology	Male	46-60	Thursday, May 09, 2019	Wednesday, May 15, 2019	6	40,540.10
John Ward	J. Fields	Gastroenterology	Female	46-60	Wednesday, May 08, 2019	Wednesday, May 15, 2019	7	12,090.53
John Ward	J. Fields	Neurology	Female	46-60	Wednesday, May 08, 2019	Wednesday, May 15, 2019	7	12,090.53
Jonathan Mcdonald	D. Rodriguez	Oncology	Female	61+	Tuesday, May 14, 2019	Wednesday, May 15, 2019	1	22,539.98
Lauren Hayes	M. Cooley	Cardiology	Female	18-30	Monday, May 13, 2019	Wednesday, May 15, 2019	2	23,423.44
Manuel Kelly	M. Jackson	Cardiology	Female	31-45	Tuesday, May 14, 2019	Wednesday, May 15, 2019	1	17,608.42
Marilyn Figueroa	A. Howard	Orthopedics	Female	61+	Tuesday, May 14, 2019	Wednesday, May 15, 2019	1	39,605.89

Figure 6: Drill-down Records page

The table includes fields such as: Patient Name, Doctor Name, Department,Gender, Age Group, Admission Date, Discharge Date, Hospital Stay Days, Treatment Cost

This page allows users to apply filters and review the detailed records behind the aggregated dashboard visuals. For example, after identifying a department with high treatment cost, managers can use this page to filter by department and inspect the corresponding patient encounter records.

5.7 Dashboard Interactivity

The dashboard includes interactive filters and slicers to support flexible analysis. The main filters include:

- Year
- Department
- Gender
- Age Group
- Doctor Name, used mainly in the drill-down page

These filters allow users to narrow the analysis based on specific decision-making needs. For example, users can filter the dashboard by year to observe performance changes over time, or by department to focus on a specific area of hospital operations.

The dashboard also supports a structured decision flow:

1. Hospital Overview — monitor overall hospital performance
2. Department Performance — compare departments by workload, cost, and stay duration
3. Root-Cause Analysis — explore possible cost and workload patterns
4. Drill-down Records — inspect detailed patient encounter records

This flow helps users move from summary-level insights to detailed investigation.

5.8 Application Relevance to the Business Problem

The developed dashboard addresses the business problem by transforming fragmented healthcare records into an interactive analytical application. Instead of manually reviewing raw records or separate spreadsheets, hospital decision makers can use the dashboard to monitor key indicators, compare departments, and explore detailed records through filters.

The application supports different user groups:

- Hospital executives can use the overview page to monitor overall performance.
- Department managers can use the department page to compare workload, cost, and length of stay.

- Operational and administrative staff can use the drill-down page to inspect detailed encounter records and support follow-up analysis.

By providing a centralized and visual interface, the dashboard improves data accessibility and supports more evidence-based decision-making in hospital performance management.

5.9 Summary

Overall, the developed Power BI dashboard provides a practical BI application for hospital performance analytics. It connects to a cleaned BigQuery analytical view and presents the data through four interactive pages: Hospital Overview, Department Performance, Root-Cause Analysis, and Drill-down Records.

The application supports hospital managers in monitoring workload, treatment cost, and length-of-stay performance. It also enables users to compare departments, explore cost and workload patterns, and inspect detailed patient encounter records. Although the current dataset limits the scope of advanced healthcare indicators, the dashboard successfully demonstrates how Business Intelligence and related technologies can transform structured healthcare data into useful decision-support insights.

6 Evaluation

6.1 Evaluation Criteria

To assess the effectiveness of the developed BI system, the following evaluation criteria are applied:

Table 4: Evaluation Criteria for the BI System

Criterion	**Description**	**Assessment**
Data Integration	Consolidate data from multiple sources into a unified warehouse	Integrated 2 Kaggle datasets into a single BigQuery star schema
Query Performance	Speed and efficiency of analytical queries	Sub-second BigQuery response; Power BI loads within 2–3s
Analytical Depth	Multi-dimensional exploration (drill-down, filtering)	4-level drill path: Overview → Department → Root-Cause → Detail
Decision Support	Ability to answer key business questions	Identify top-cost departments, compare ALOS within 2–3 clicks
Usability	Ease of use for non-technical users	Interactive slicers, KPI cards, structured navigation
Scalability	Handle growing data volumes	Cloud-based BigQuery scales without infrastructure changes

6.2 Achievements

The project successfully demonstrates the following:

1. **Complete BI Pipeline Implementation:** The system covers the full data lifecycle—from raw data extraction through ETL processing, data warehousing (star schema on BigQuery), to interactive visualization (Power BI). This demonstrates a practical, end-to-end application of concepts from the Data Warehouses and Decision Support Systems course.

2. **Effective Star Schema Design:** The dimensional model follows Kimball methodology with a clear fact table (patient encounters), patient, doctor, department, and date dimensions, with a clear encounter-level fact table.

3. **Interactive Multi-page Dashboard:** The Power BI application provides four distinct analytical views, each serving a different level of decision-making:

- Hospital Overview: High-level KPI monitoring
- Department Performance: Comparative departmental analysis
- Root-Cause Analysis: Pattern exploration across demographics and departments
- Drill-down Records: Detailed patient encounter inspection

4. **Support for Multiple Stakeholders:** The dashboard design accommodates different user roles—from hospital executives needing summary KPIs to operational staff requiring detailed record access.

5. **Cloud-based Architecture:** Using Google BigQuery as the data warehouse demonstrates modern cloud-based BI practices, providing scalability and eliminating infrastructure management overhead.

6.3 Limitations

Despite the achievements, the current implementation has several limitations:

1. **Simulated Data:** The system uses publicly available Kaggle datasets rather than real hospital data. While the data structure is representative, it may not capture the full complexity of real-world healthcare operations (e.g., missing fields for diagnosis codes, medication details, readmission tracking).

2. **Limited KPI Coverage:** Due to dataset constraints, several important healthcare KPIs could not be implemented:

   - Readmission Rate (no readmission tracking in the dataset)
   - Bed Occupancy Rate (no bed utilization data available)
   - Patient Satisfaction Score (no feedback/survey data)
   - Mortality Rate or Clinical Outcome metrics

3. **No Real-time Data Integration:** The current ETL process is batch-based (one-time load). A production system would require scheduled or real-time data pipelines to keep the warehouse current.

4. **Limited Predictive Capabilities:** The system focuses on descriptive analytics (what happened) and basic diagnostic analytics (why it happened). Advanced predictive analytics (e.g., forecasting patient volumes, predicting readmission risk) are not implemented.

5. **Single Data Warehouse View:** The Power BI dashboard connects to a single analytical view. A more mature implementation would leverage multiple views or materialized tables optimized for different analytical scenarios.

6.4 Comparison with Project Goals

Table 5: Goal Achievement Assessment

Goal	**Status**	**Notes**
Build an integrated Data Warehouse	Achieved	Star schema on BigQuery with multiple dimension tables
Develop interactive BI Dashboard	Achieved	4-page Power BI dashboard with slicers and drill-downs
Enable Multi-dimensional Analysis	Achieved	Filtering by time, department, doctor, demographics
Support Root-Cause Investigation	Partial	Supported for available dimensions; limited by dataset
Demonstrate Complete BI Pipeline	Achieved	ETL $\rightarrow$ DW $\rightarrow$ Visualization fully implemented

7 Conclusion

7.1 Summary

This project successfully demonstrates the application of Business Intelligence and related technologies to the domain of Hospital Performance Analytics. By implementing a complete data pipeline—from ETL processing with Python/Pandas, through dimensional modeling with a star schema on Google BigQuery, to interactive visualization with Power BI—the project illustrates how modern BI/DSS technologies can transform fragmented healthcare data into actionable decision-support insights.

The developed system addresses the core business problem of data fragmentation and lack of analytical capability in hospital management. Through an interactive 4-page dashboard, hospital decision-makers gain the ability to:

- Monitor key performance indicators (Total Encounters, ALOS, Treatment Cost) at a glance.

- Compare departmental performance across multiple metrics.
- Investigate root causes behind cost and workload patterns.
- Access detailed patient encounter records for follow-up analysis.

7.2 Lessons Learned

Through this project, the team gained practical experience in:

1. **Dimensional Modeling:** Designing a star schema that balances query performance with analytical flexibility, including role-playing date dimensions and clear fact-dimension relationships.
2. **ETL Development:** Building data transformation pipelines that handle real-world data quality issues (inconsistent formatting, missing values, derived calculations).
3. **Cloud Data Warehousing:** Working with Google BigQuery as a serverless analytical platform, understanding its strengths for OLAP workloads.
4. **BI Dashboard Design:** Creating user-centered dashboards that support a logical decision-making flow from summary to detail.
5. **Integration of BI and DSS Concepts:** Understanding how BI (monitoring, reporting) and DSS (decision support, root-cause analysis) complement each other in a unified system.

7.3 Future Work

To extend this project into a more comprehensive hospital analytics platform, the following improvements are recommended:

1. **Integration of Additional Data Sources:** Incorporate real EHR data, patient satisfaction surveys, bed utilization records, and diagnosis/ICD-10 codes to enable a broader set of healthcare KPIs.
2. **Automated ETL Pipeline:** Implement scheduled data pipelines (e.g., using Apache Airflow or Google Cloud Dataflow) to support periodic or near-real-time data refresh.
3. **Predictive Analytics:** Add machine learning models for forecasting patient volumes, predicting readmission risk, or estimating resource requirements.

4. **Advanced DSS Features:** Implement what-if analysis capabilities (e.g., "What happens to bed occupancy if we add 10 beds to the Cardiology department?") and scenario comparison tools.

5. **Role-based Access Control:** Implement row-level security in Power BI so that department heads only see data relevant to their departments while executives retain full visibility.

6. **Mobile and Real-time Dashboards:** Extend the dashboard to support mobile access and real-time data streaming for critical KPIs that require immediate attention.

8 References

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[5] Ralph H. Sprague. A framework for the development of decision support systems. *MIS Quarterly*, 4(4):1–26, 1980.

[6] Efraim Turban, Ramesh Sharda, and Dursun Delen. *Decision Support and Business Intelligence Systems*. Prentice Hall, 9th edition, 2011.

Appendix 1. Teamwork report

No.	Full Name	Assigned Work	Status
1	Nguyễn Phúc Thanh Danh	Introduction, business problem, evaluation, conclusion	100 %
2	Phạm Viết Hào	Extract - Transform - Load	100 %
3	Phạm Thanh Phong	Application Development	100 %
4	Trần Ngọc Thoại	Proposed solution	100 %

Appendix 2. Dashboard File

The Power BI dashboard file is uploaded as: **Hospital_Performance_Dashboard.pbix** via the link: https://drive.google.com/drive/folders/1FX78ZvhG2ojf-1JmnQUV6vJhjQly7gpx?usp=sharing

This file contains the implemented interactive dashboard with four pages.